FloodOpt

" ?4=-B>DA24 ?;0C5>A< E>>A >?C8<0;8BOC84 E0= 389:E4ABC4A:8=6 4= A8E84AE4AAD8<8=6.

Doel

F;>>3 " ?C >?C8<0;8B44AC F0C4AE48;8674 83BBCA0C468 4J= >? CA0942C-, 389:A8=6- 4= BHBC44<=8E40D 3>>A 389:E4ABC4A:8=6 4= A8E84AE4AAD8<8=6 C4 2><18=4A4=. H4C ?;0C5>A< >=34ABC4D=C ?A>1018;8BC8B274 A8B82>'B, <44A34A4 :;8<00CB24=0A8>'B, <44A34A4 500;<4270=8B<4= 4= A>1DDBC4 >?C8<0;8BOC84.

'>4:><BC864 C>4?0BB8=64=: H * B#-0=0;HB4B, KBA, 0BB4C <0=064<4=C, :;8<00CBCA4BBC4BC4= E>>A 58=0=28J;4 8=BC4;;8=64=, ?>AC454D8;;4->?C8<0;8BOC84.

Architectuur

```
cddddddddddddddddddddddddddddddddddddddddddddddddddddddb
]  F+('-e'd  Reac- V V"-e V Leaf%e-                     ]
]  Kaa+-e'7 Ge(JSON /a' API  ;  Re,.%-a-e'9da,!b(a+d    ]
`ddddddddddddddddddddaddddddddddddddddddddddddddddddddd_
                     ] HTTP : Ge(JSON
cdddddddddddddddddddddZddddddddddddddddddddddddddddddddb
]  Fa,-API  >d.''e HTTP9,c!"%6  ee' b.,"'e,, %( "c?    ]
]  POST :()-"&"3e  \  f%((d()-9c(+e                     ]
]  GET  :-+a#ec-(+"e,:@"dA: e(#,('  \  Ge(Pa'da,       ]
`ddddddaddddddddddddddddddddddddddddddddddaddddddddddddd_
       ] SQLA%c!e&2                      ] Ge(Pa'da,
cddddddZddddddddddb          cddddddddddddZddddddddddddb
]  SQL"-e >de/?   ]          ]  Ge(&e-+"e             ]
]  P(,- +eSQL     ]          ]  WKT : Ge(JSON         ]
]  >)+(d6 ()-8?   ]          ]  ,!a)ef"%e,            ]
`ddddddddddddddddd_          `ddddddddddddddddddddddddd_
                     ]
cdddddddddddddddddddddZddddddddddddddddddddddddddddddddb
]  f%((d()-9c(+e  >,-+"$-e %aa ,c!e"d"' ?               ]
]                                                       ]
]  O)-"&"3a-"(' La2e+  &"' Cce1e  ,8-8 C!e1e Y !J&"' ]
]         [  +(e)- aa'                                  ]
]  R",$ La2e+      NCW X C P>,?=VU=eH>>EWF?,?          ]
]         [  +(e)- aa'                                  ]
]  P!2,"c, La2e+   P>-? X PU=eH>DG-?=eH>WDB!?          ]
`ddddddddddddddddddddddddddddddddddddddddddddddddddddddd_
```

,84 `d(c,:a+c!"-ec-.+e8)'` E>>A 74C E>;;43864 0A278C42CDDA3806A0<.

Tech stack

Component	Technologie	Reden
%4:4=:4A=4;	#HC7>= (`f%((d()-9c(+e`)	#DA4 #HC7>=, 644= 5A0<4F>A:- ;>2:-8=

Component	Technologie	Reden
" ?C8<8I4A	#H><> 6 + H8GH& (IL#)	" ?4=-B>DA24, B2700;C =00A 6A>C4 !
B02:4=3 A#I	F0BCA#I	&=4;, 0DC><0C8B274 &F0664A (I
D0C010B4	**SQLite** (34E) M #>BC6A4&$L >?C8>=44; (?A>3)	! D; 8=BC0;;0C84 E>>A 34E4;>?<4=C
G4>-E4AF4A:8=6	**GeoPandas** + G4>J&" !	&4AE4A-B834 64><4CA84 I>=34A #>BCGI&
K00AC4= (5A>=C4=3)	**Leaflet** + %402C +)8C4	L44BC G4>J&" ! 38A42C E0= A#I
$D4D4	%438B + C4;4AH *(stap 2.3)*	ABH=2 >?C8<0;8B0C84B
D>2D<4=C0C84	<0C?;>C;81 (L0'4+-5>A<D;4B)	

`d(c$e+9c(&)(,e82&%` 4= `"'"-J,c!e&a>?` BC00= 0; :;00A.

Projectstructuur

```
f%((d()-:
^dd f%((d()-9c(+e:           L +e$e'$e+'e% <  ee' Fa,-API6 ).+e P2-!('
]   `dd f%((d()-Jc(+e:
]       ^dd "(:              L P2da'-"c da-a&(de%%e' >Mea,.+e6 Sce'a+"(6 T+a#ec-(+2?
]       ^dd )!2,"c,:         L P!2,"c,M(de% P+(-(c(% V S"&)%eD"$eO/e+f%(0
]       ^dd +",$:            L R",$Ca%c.%a-(+ P+(-(c(% V S"&)%eR",$Ca%c.%a-(+
]       ^dd ()-"&"3a-"(':    L O)-"&"3a-"('S-+a-e 2 P+(-(c(% V B+.-eF(+ce V P2(&(
]       `dd .-"%,:
^dd f%((d()-9a)":            L Fa,-API bac$e'd
]   `dd f%((d()-Ja)":
]       ^dd &a"'8)2          L e'd)("'-, V DI
]       ^dd da-aba,e8)2      L SQLA%c!e&2 ORM >SQL"-e : P(,- +eSQL?
]       `dd +e)(,"-(+"e,8)2  L Me&(+2 V P(,- +e, "&)%e&e'-a-"e,
^dd f%((d()-90(+$e+:         L Ce%e+2 0(+$e+, >,-a) 08P?
^dd f%((d()-9f+('-e'd:       L Reac- V V"-e V Leaf%e- >,-a) Q?
^dd -e,-,:
]   ^dd .'"-:                L QS -e,-, < a%%e %a e'
]   ^dd "'-e +a-"(':         L PT -e,-, < CLI6 API6 da-aba,e
]   `dd /a%"da-"(':          L ()-"&a%",eJ+"' JOMNN8,*%"-e >+efe+e'-"eda-a?
^dd d(c,:                    L PNG9d"a +a&&e' >/"a  e'e+a-eJd(c,8)2? V SVG b+(''e'
`dd ,c+")-,:                 L  e'e+a-eJd(c,8)26 +.'J,&($eJ-e,-8)26 "'"-Jdb8)2
```

Ontwikkelprincipes

- **Strikte laagscheiding** L >?C8<8I4A 14EOC =>>8C 5HB8B274 5>A<D;4B; A#I 14EOC 644= 1DB8=4BB ;>682
- **Modulaire interfaces** L 4;:4 ;006 8<?;4<4=C44AC 44= #A>C>2>;; 8<?;4<4=COC84B I89= E4AE0=6100A
- **Dubbele validatie** L 4;:4 >?C8<0;8BOC84 64E4A858 44A3 <4C 1ADC4-5>A24 A454A4=C84
- **Installatie-arm** L &$L8C4 + 8=-<4<>AH C4BCB; 644= D>2:4A E4A48BC E>>A 34E4;>?<4=C
- **Stapsgewijze uitbreiding** L) # 44ABC, #>BCGI& / C4;4AH / 5A>=C4=3 ?0B 0;B =>386

Documentatie

A;;4 3806A0< <4= F>A34= 6464=4A44A3 <4C `)2-!(' ,c+")-,: e'e+a-eJd(c,8)2`:

Diagram	Bestand
)>;;43864 0A2784C42CDDA (A4:4=:4A=4; + A#I + 64> + 5A>=C4=3)	`d(c,:a+c!"-ec-.+e8)'`
#7HB82B L0H4A L P (t) 5>A<D;4 + 2DAE4B	`d(c,:,-a)N8NJ)!2,"c,Jf(+&.%a8)'`
%8B: L0H4A L !C * 14A4:4=8=6	`d(c,:,-a)N8OJ+",$J'c08)'`
"?C8<8I0C8>= L0H4A L BADC4F>A24 EB #H><>	`d(c,:,-a)N8PJ()-"&"3a-"('8)'`
&<>:4 C4BC L 4=3-C>-4=3 E4A858C20C84	`d(c,:,-a)N8QJ,&($eJ-e,-8)'`
F0BCA#I B4AE824 L 4=3?>8=CB + 5;>F	`d(c,:,-a)O8NJa)"8)'`
D0C010B4 L &$L8C4/#>BC6A4&$L + A4?>B8C>AH-?0CC4A=	`d(c,:,-a)O8OJda-aba,e8)'`
G4>-BC02: L G4>#0=30B + L405;4C + G4>J&" !	`d(c,: e(J,-ac$8)'`
D0C010B4 <0??8=6 DB M F;>>3"?C	`d(c,:da-aba,eJ&a))"' 8)'`

Fasering

Fase	Status	Inhoud
0 L '>>;8=6	✅ K;00A	#A>942CBCAD2CDDA, ?02:068=6, ?A4-2>< <8C
1 L)# A4:4=:4A=4;	✅ K;00A	#7HB82B, %8B:, "?C8<8I0C8>=, CLI B<>:4 C4BC
2 L B02:4=3 & A#I	🔄 2.2 :;00A	F0BCA#I ✅, &$L8C4 ✅, C4;4AH ⏳
3 L (8C1A4838=64= A4:4=:4A=4;	⏳ G4?;0=3	F"% / >=C4 C0A;>, ;4=6C4-45542C4=, A8E84AE4AAD8<8=6
4 L FA>=C4=3	⏳	%402C +)8C4 + L405;4C, G4>#0=30B G4>J&" !

Stap 0.2 ✅ — Data model

#H30=C82 E2 <>34;B: `Mea,.+e` (45542C -<.), `Sce'a+"(` (4C0 -</9.), `T+a#ec-(+2` (?0, 0;?70, 10B4/H40A).

Fase 1 — MVP rekenkernel (traject-niveau)

Stap 1.1 ✅ — Physics Layer

$$P(t) = P_0 \cdot e^{\alpha \eta t} \cdot e^{-\alpha \Delta h}$$

`S"&)%eD"$eO/e+f%(0` 027C4A `P!2,"c,M(de%` #A>C>2>;. F>A<D;4 834=C84: 00= "?C8<0;8B4%8=6 2.3.2 (HK), 2013).

Stap 1.2 ✅ — Risk Layer

$$\mathrm{NCW} = \sum_{s=0}^{T-1} P(s) \cdot V_0 \cdot e^{(\gamma-\delta)s}$$

`S"&)%eR",$Ca%c.%a-(+` 027C4A `R",$Ca%c.%a-(+` #A>C>2>;.

Stap 1.3 ✅ — Optimization Layer

Objective	Formulering	Solver
`MINJCOST`	$\min \sum c_i x_i$ s.t. $\sum h_i x_i \geq h_{\min}$	H8GH& IL# (4G02C)
`MAXJRISKJREDUCTION`	$\max \sum h_i x_i$ s.t. $\sum c_i x_i \leq B$	H8GH& IL# (4G02C)
`MINJNCW`	$\min \sum (c_i - C\alpha h_i) x_i$	H8GH& IL# (;8=408A4 14=034A8=6)

`B+.-eF(+ce8,(%/e>? XX P2(&(O)-"&"3e+8,(%/e>?` E>>A 0;;4 6 C4BC20B4B ✅

Stap 1.4 ✅ — Integratie smoke test (CLI)

```
)2-!(' ,c+")-,:+.'J,&($eJ-e,-8)2
```

Fase 2 — Backend & API

Stap 2.1 ✅ — FastAPI service

```
POST :,ce'a+"(,        POST :-+a#ec-(+"e,
POST :()-"&"3e         GET  :+e,.%-,:@#(bJ"dA
GET  :d(c,             >S0a  e+ UI?
```

Stap 2.2 ✅ — Database

SQLite (standaard — nul installatie):

```
./"c(+' f%((d()-Ja)"8&a"'7a)) 99+e%(ad
L \ ,c!+"#f- 'aa+ f%((d()-8db
```

PostgreSQL (productie — optioneel):

```
d(c$e+ c(&)(,e .) 9d )(,- +e, 55 )2-!(' ,c+")-,:"'"-Jdb8)2
DATABASEJURLX)(,- +e,*%7::f%((d()-7f%((d()-K%(ca%!(,-7RQPO:f%((d()- I
  ./"c(+' f%((d()-Ja)"8&a"'7a)) 99+e%(ad
```

&274<0: `,ce'a+"(,`, `-+a#ec-(+"e,`, `()-"&"3a-"('J+e,.%-,`.
#>BCGI&-4GC4=B84 00=64<00:C 189 #>BC6A4&$L (E>>A 64><4CA84 BCO? 4.2).

Stap 2.3 ⏳ — Async queue (Redis + Celery)

`POST :()-"&"3e` 6445C 38A42C `#(bJ"d` C4AD6. &COCDB: `)e'd"'  \ +.''"'  \ d('e`.
&$L8C4 M #>BC6A4&$L BF8C27 E4A?;827C 189 <44A34A4 C4;4AH-F>A:4AB (2>=2DAA4=C FA8C4B).

Fase 3 — Uitbreidingen rekenkernel ⏳

Stap	Inhoud	Wat blijft ongewijzigd
3.1	#A>101;8B8B274 BC4A:C4: F"% / >=C4 C0A;>	"?C8<8I4A, %8B: L0H4A
3.2	L4=6C4-45542C4= & 2>AA4;0C84B	"?C8<8I4A, #7HB82B L0H4A
3.3	%8E84AE4AAD8<8=6 & 7H3A0D;8B274 8=C4A02C84B	"?C8<8I4A, %8B: L0H4A

!84DF4 8<?;4<4=C0C84B 027C4A 14BC00=34 #A>C>2>;B L >?C8<8I4A 7>45C =84C 00=64?0BC C4 F>A34=.

Fase 4 — Frontend ⏳

React + Vite + Leaflet

Stap 4.1 — Frontend setup

%402C +)8C4, L405;4C E>>A 8=C4A02C84E4 :00AC4=, '0=&C02: $D4AH E>>A A#I-20;;B.

Stap 4.2 — Kaartviewer (GeoPandas + Leaflet)

D89:E0:-64><4CA84 (* K ' 8= &$L8C4) M G4>#0=30B M G4>J&" ! M L405;4C:

```
API e'd)("'-7  GET :-+a#ec-(+"e,:@"dA: e(#,('
Se+/e+7        Ge(Pa'da, %ee,- WKT6  eef- Ge(JSON -e+.
F+('-e'd7      Leaf%e- +e'de+- d"#$/a$$e' () $aa+-
               K%e.+ +ad"4'- () ba,", /a' P>-? (f NCW
```

Stap 4.3 — Scenario-editor & resultaten-dashboard

00CA464;-438C>A, B24=0A8>-B4;42C84, !C * -6A0584:4=, E4A64;89:8=6 BADC4F>A24 EB #H><>.

Licentie

" ?4= B>DA24 L ;824=C84 E>;6C.